

TEORIA DA EVOLUÇÃO DAS ESPÉCIES: SELEÇÃO NATURAL

Charles Darwin

1859



VARIABILITY



Darwin's Finches
(*Geospiza*)

SELECTION
(the struggle for existence)



HEREDITY



Santa Cruz Island

ADAPTATION
(*Geochelone*)



ADAPTATION

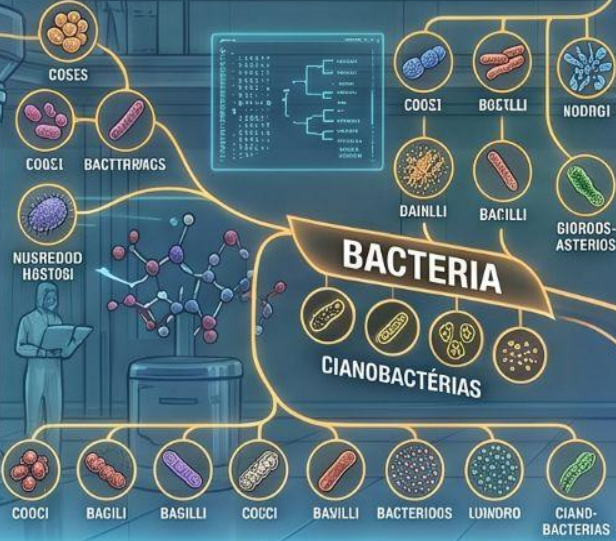


ÁRVORE DA VIDA

PLANTAS



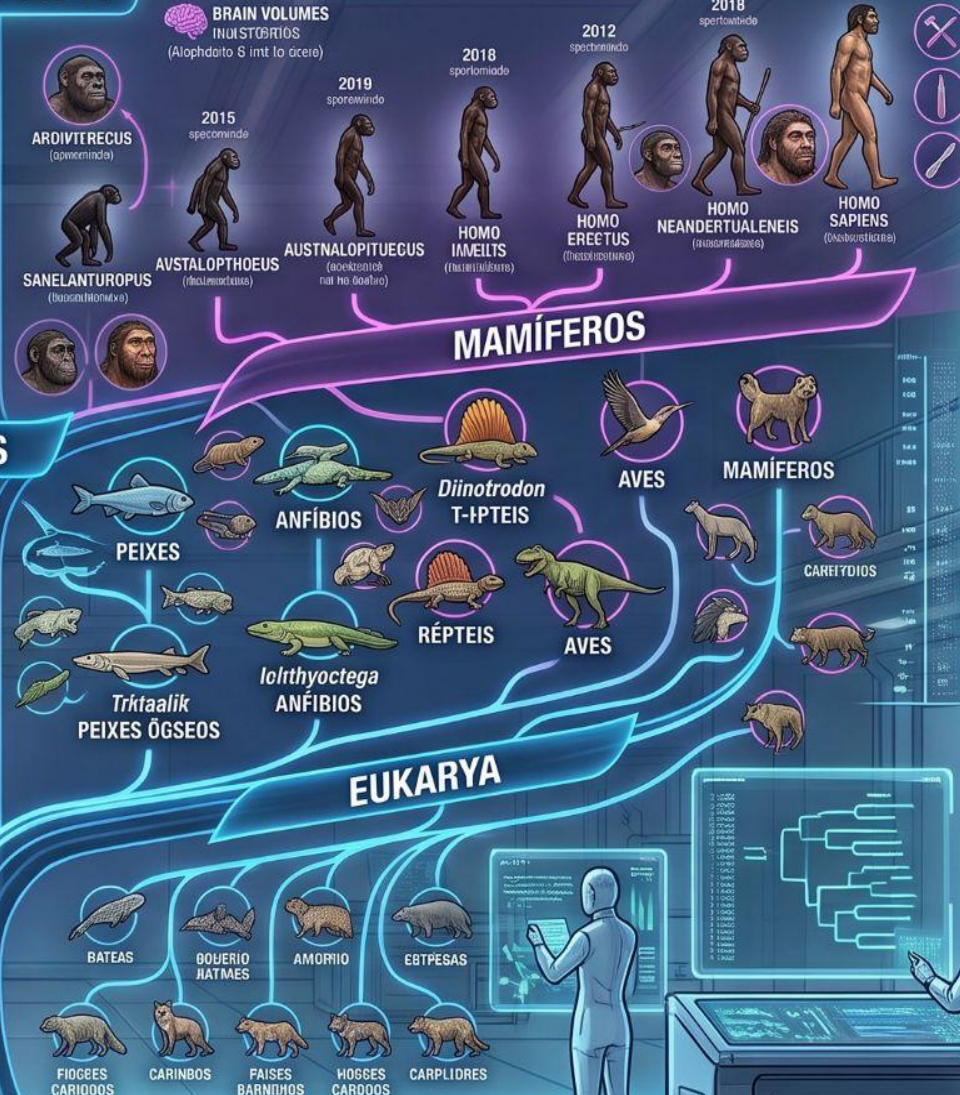
PLANTAS



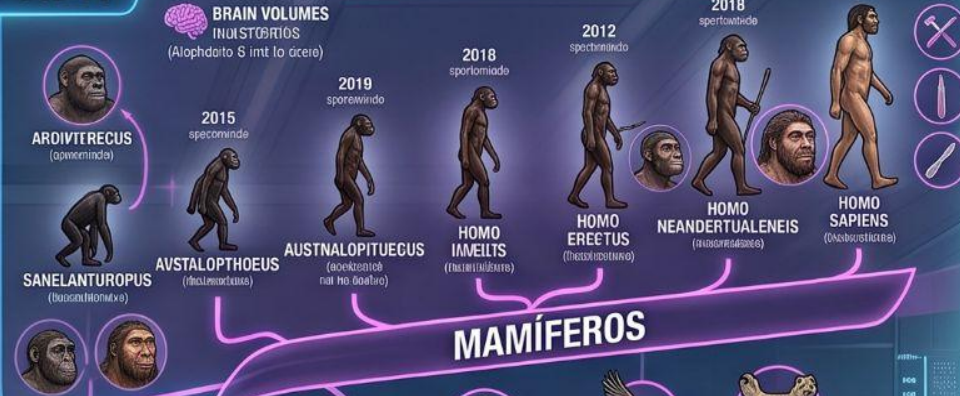
BACTERIA

CIANOBACTERIAS

ANIMAIS



EVOLUÇÃO DE HOMINÍDEOS



MAMÍFEROS

EUKARYA

LUCA Ancestral Comum

1866

EXPERIMENTO DE MENDEL

FLOR

Posição

Cor

Altura

Textura

Cor

VAGEM

Forma

Cor



Axial

Branca

Alta

Lisa

Amarela

Inflada

Amarela



Terminal

Púrpura

Baixa

Rugosa
toda matéria

Verde

Comprimida

Verde

Gregor Mendel "Fenótipo"



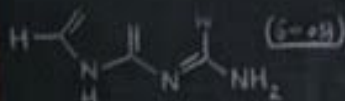
**Friedrich
Miescher DNA**

1877



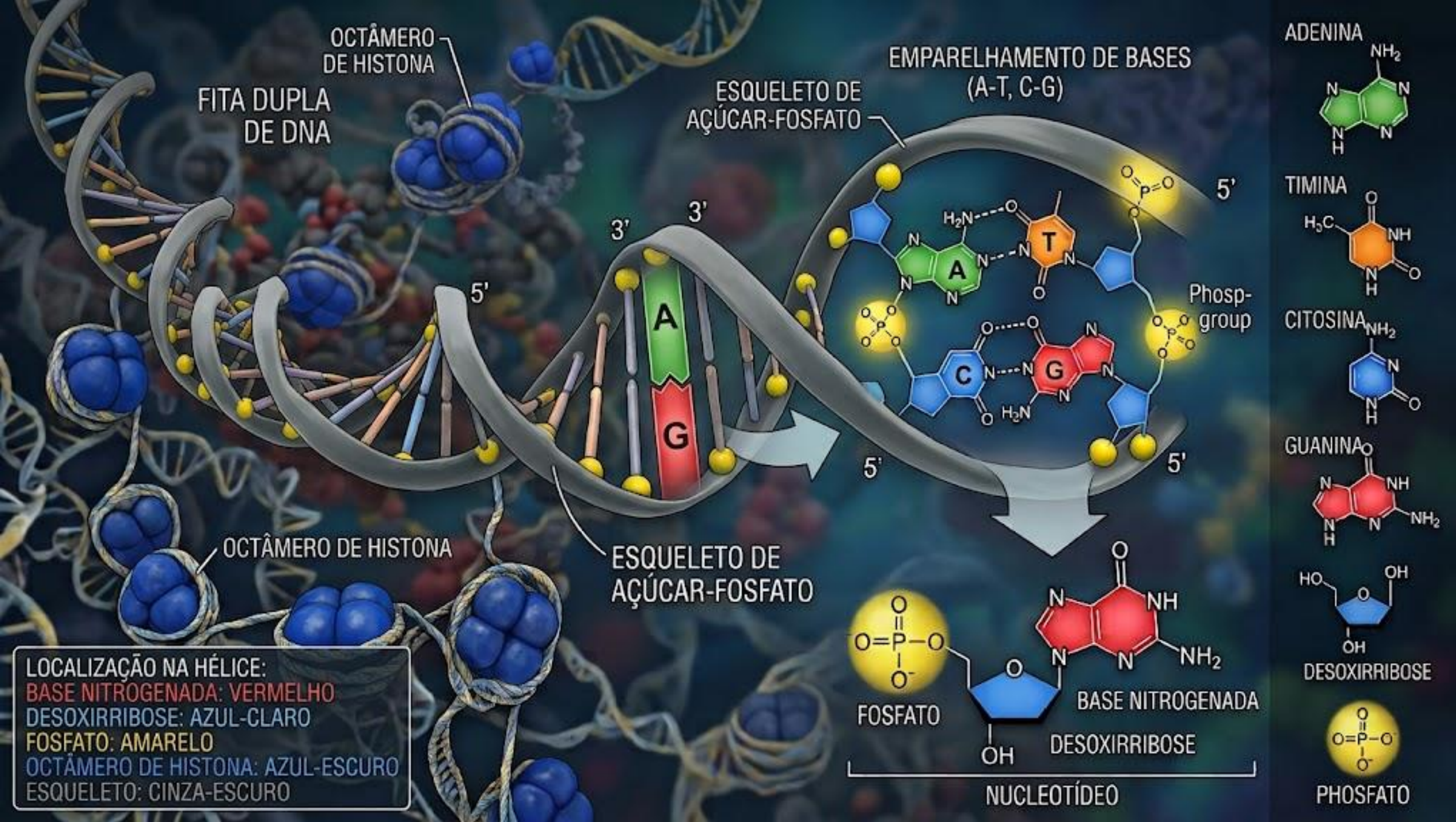
James Watson
& Francis Crick

DNA = Double Helix
Adenine = Thymine
Guanine = Cytosine



The Double Helix, 1953

Crick & Watson,
Nature 1953

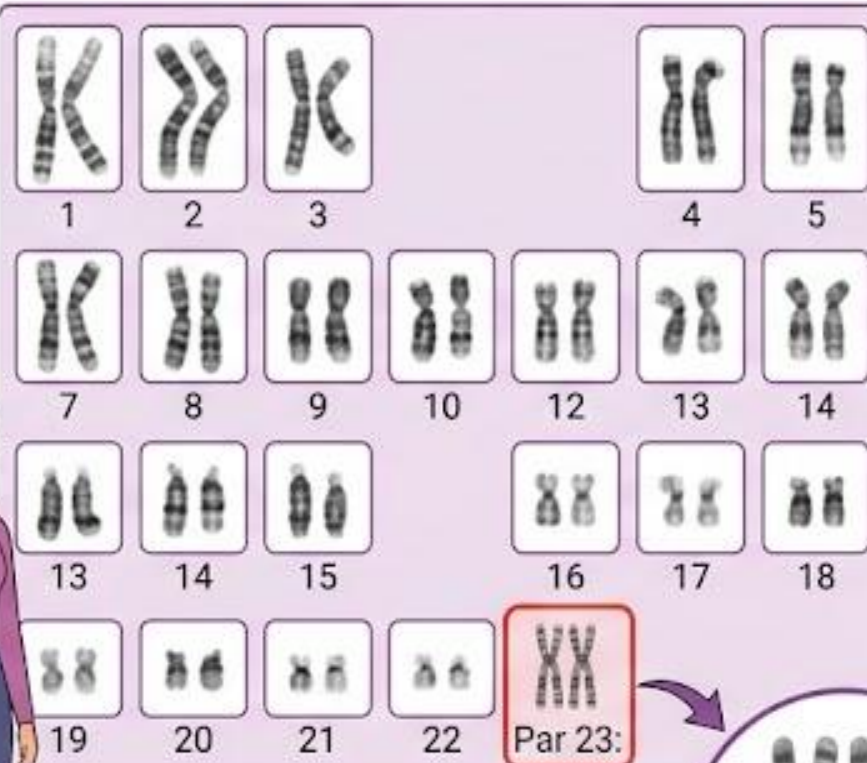


CARIÓTIPO HUMANO: OS 23 PARES DE CROMOSSOMOS E DETERMINAÇÃO SEXUAL



CARIÓTIPO FEMININO (XX)

Pares Autossômicos



XX: FEMININO
(Par 23, idênticos)

XX: FEMININO
Sexuais

Par 23 destacado em **vermelho**
para evidenciar a diferença



TOTAL: 46
CROMOSSOMOS
(23 PARES)

22 PARES
AUTOSSÔMICOS:
Comuns a ambos



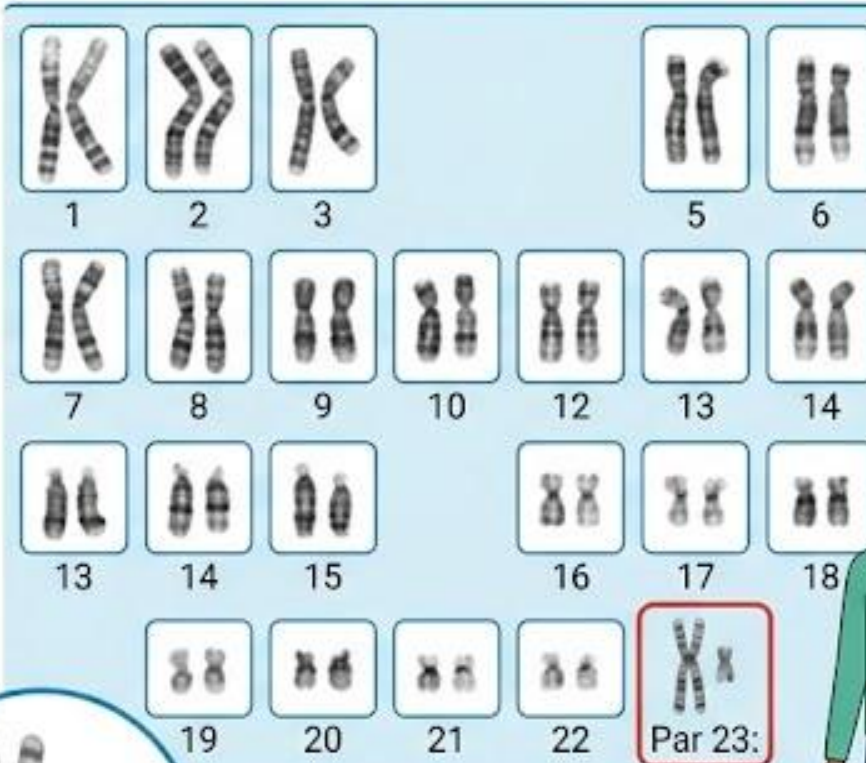
1 PAR SEXUAL:
Determina o sexo
(XX ou XY)



CARIÓTIPO MASCULINO (XY)



Pares Autossômicos



XY: MASCULINO
(Par 23, diferentes)

XY: MASCULINO
Sexuais

Par 23 destacado em **vermelho**
para evidenciar a diferença



ANTES DA MUTAÇÃO

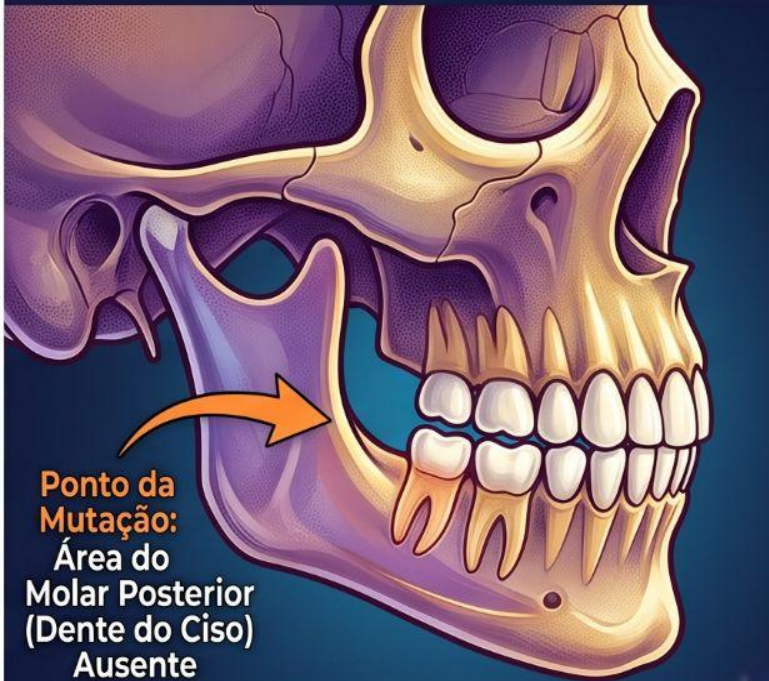


**PROCESSO
DE MUTAÇÃO**

DEPOIS DA MUTAÇÃO

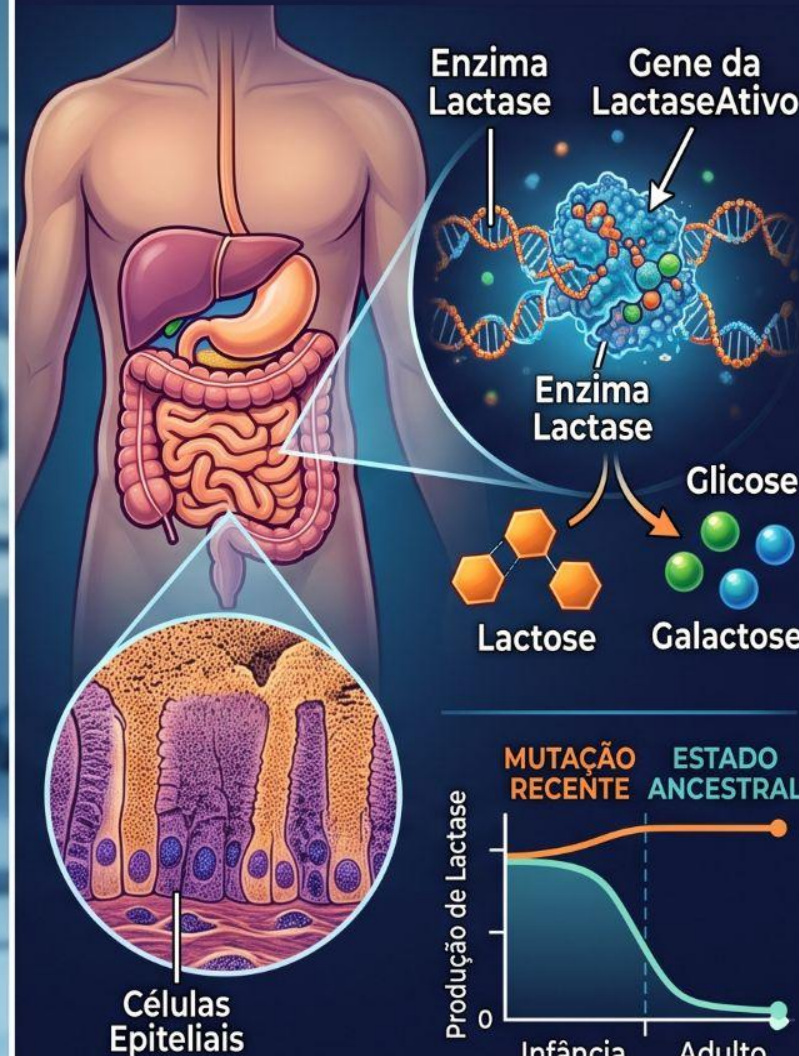


1. AUSÊNCIA DE DENTES DO CISO



Mandíbula de Ancestral (com ciso)

2. TOLERÂNCIA À LACTOSE NA IDADE ADULTA



3. PERSISTÊNCIA DA ARTÉRIA MEDIANA



1975



LEDÉ

ERNŐ RUBIK - CREATOR

ERNŐ RUBIK
- CREATOR

BÜVÖS
KOCKA

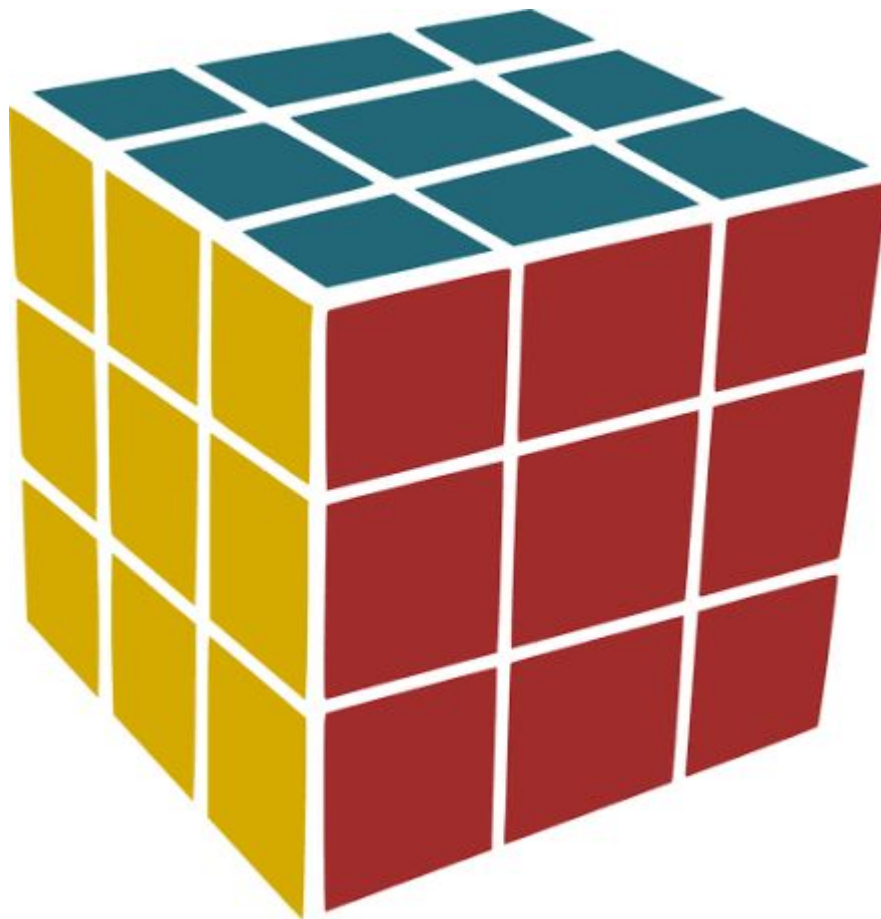


1. VILÁGBAJNOKSÁG - BÚVÖS KOCKA BUDAPEST 1982

OFFICIAL RESULT		
1	NG THAI (USA)	23.94 (NEW)
2	J. OYONGY (HUN)	
3		
4	JESSICA FRIDRICH (TEX)	28.11
5		
6		
7		
8		
9		
10		
11		
12		

RECORD TIME:
22.95
SECONDS

POSIÇÃO:
10º LUGAR
29.11s (JESSICA)



CFOP: JESSICA FRIDRICH METHOD



CFOP: O MÉTODO MAIS POPULAR DO MUNDO - JESSICA FRIDRICH

1



C (CROSS)

FORME A CRUZ
NA FACE BRANCA



2



F (F2L)

RESOLVA AS DUAS
PRIMEIRAS CAMADAS
JUNTAS

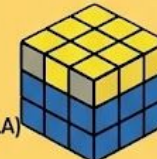


3



O (OLL)

ORIENTE A ÚLTIMA
CAMADA (FACE AMARELA)



4



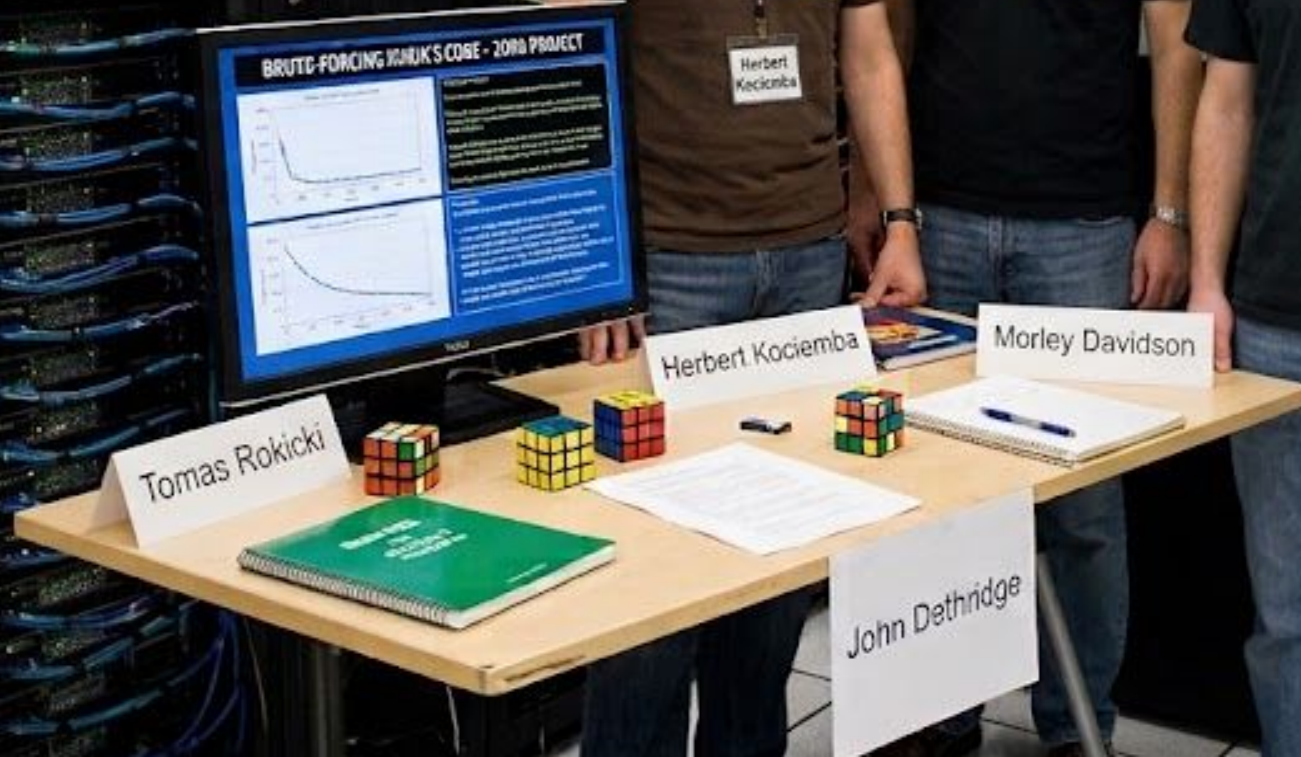
P (PLL)

PERMUTE AS PECAS
PARA FINALIZAR O CUBO



COMECE A APRENDER

2008





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